



GLA
UNIVERSITY
Recognized by UGC Under Section 2(f) & 12B Status

Accredited with **A+** Grade by **NAAC**

Mathura | Greater Noida



OFFICE OF STUDENTS' WELFARE

DEPARTMENT OF COMPUTER ENGINEERING AND APPLICATION

GLA UNIVERSITY | CYBERONITES

P R E S E N T S

INTRUSION

SECOND EDITION

PROBLEM STATEMENTS



@official_cyberonites

INTRUSION

SECOND EDITION

ARTIFICIAL INTELLEGEENCE



@official_cyberonites

AI/ML

1. AI-Based Malware Classification System

Problem Statement:

Malware attacks are evolving rapidly, making traditional signature-based detection systems ineffective.

Description:

Modern malware variants constantly change their signatures to evade traditional antivirus systems. AI-based malware classification systems can analyze behavioral patterns, file structures, and execution traces to identify malicious software.

Challenges:

- Detecting polymorphic and obfuscated malware.
- Handling large malware datasets efficiently.
- Reducing false positives while maintaining detection accuracy.

INTRUSION
SECOND EDITION



AI/ML

2. AI-Based Document Forgery Detection

Problem Statement:

Forged documents and identity fraud are increasing across financial institutions, government services, and digital verification systems.

Description:

Fraudsters manipulate digital documents such as ID cards, certificates, and invoices. AI-based systems can analyze image inconsistencies, text anomalies, and metadata to detect forged documents.

These systems may use computer vision, pattern recognition, and machine learning techniques to identify alterations such as tampered text, inconsistent fonts, altered signatures, or manipulated images in scanned documents.

Challenges:

- Detecting subtle manipulations in scanned documents.
- Handling multiple document formats and layouts.
- Ensuring high detection accuracy for real-world use cases.

INTRUSION
SECOND EDITION



AI/ML

3. AI-Based Insider Threat Detection

Problem Statement:

Organizations face serious security risks from insider threats where employees misuse access privileges to leak sensitive information.

Description:

Insider threats are difficult to detect because malicious activities often appear as legitimate system behavior. Employees with authorized access may intentionally or unintentionally misuse sensitive data. AI-based systems can analyze user activity logs, access patterns, system behavior, and login histories to identify unusual actions that may indicate insider threats. Machine learning models can learn normal user behavior and flag anomalies such as abnormal file access, unusual login times, or large data transfers.

Challenges:

- Identifying abnormal behavior among legitimate users.
- Analyzing large volumes of user activity logs.
- Minimizing false alerts while maintaining high detection accuracy.

INTRUSION
SECOND EDITION



AI/ML

4. AI-Powered Data Privacy Compliance Checker

Problem Statement:

Organizations face difficulties ensuring compliance with data protection regulations while handling large volumes of sensitive user data.

Description:

With the increasing importance of privacy laws and regulations, organizations must ensure that personal data is processed, stored, and shared securely. However, monitoring large datasets manually for compliance violations is extremely difficult.

An AI-powered compliance checker can automatically analyze databases, logs, and storage systems to detect exposure of sensitive information, identify policy violations, and ensure adherence to data protection standards such as GDPR or other privacy frameworks.

Challenges:

- Identifying personally identifiable information (PII) in large datasets.
- Detecting unauthorized access or data leaks.
- Ensuring compliance without affecting operational efficiency.

INTRUSION
SECOND EDITION



AI/ML

5. Adversarial Attack Detection for AI Models

Problem Statement:

Develop a system that detects adversarial inputs designed to manipulate AI models.

Description:

Machine learning models can be manipulated using adversarial examples—inputs that are slightly modified in ways that are often imperceptible to humans but cause the AI model to make incorrect predictions. These attacks can compromise systems used in areas such as autonomous driving, facial recognition, and cybersecurity.

The proposed system should analyze incoming data and detect adversarial manipulations before they influence the AI model's decision-making process, helping to improve the reliability and security of AI systems.

Challenges:

- Identifying adversarial inputs that contain very small or hidden perturbations.
- Ensuring detection works across different types of AI models.
- Protecting AI systems from manipulation without affecting model performance.

INTRUSION
SECOND EDITION



AI/ML

6. AI Model Integrity Verification

Problem Statement:

Create a mechanism to verify that deployed AI models have not been tampered with.

Description:

AI models deployed in production environments may be modified or corrupted by attackers. Such tampering can lead to incorrect predictions, biased results, or malicious behavior that may compromise system reliability. The goal is to design a verification mechanism that checks the integrity of AI models before execution. This system can use techniques such as hashing, digital signatures, or blockchain-based verification to ensure that the model being used is exactly the same as the original trusted version.

Challenges:

- Detecting unauthorized modifications to AI models.
- Ensuring verification does not significantly increase system overhead.
- Maintaining secure storage and validation of model integrity data.

INTRUSION
SECOND EDITION



AI/ML

7. Secure Federated Learning Framework

Problem Statement:

Design a federated learning system that protects data privacy during distributed training.

Description:

Federated learning enables multiple devices or organizations to collaboratively train machine learning models without sharing their raw data. Instead of sending data to a central server, each participant trains the model locally and shares only the model updates. While this approach improves privacy, it introduces new security risks such as malicious participants injecting harmful updates or attempting to infer sensitive information from shared parameters. The goal is to build a secure federated learning framework that ensures privacy, protects against malicious participants, and maintains reliable collaborative training.

Challenges:

- Preventing data leakage during distributed training.
- Detecting malicious or poisoned model updates from participants.
- Maintaining model accuracy while ensuring strong privacy protection.

INTRUSION
SECOND EDITION

AI/ML

8. Poisoned Dataset Detection

Problem Statement:

Design a system that identifies malicious or poisoned data within machine learning datasets.

Description:

Machine learning models rely heavily on training datasets. If attackers inject malicious or manipulated data into these datasets, the model's behavior can be intentionally altered. This type of attack is known as data poisoning and can lead to incorrect predictions or system vulnerabilities.

The goal is to build a detection system that analyzes datasets to identify suspicious or manipulated data points before they are used in model training. The system should automatically flag or remove potentially poisoned samples to ensure the integrity and reliability of the training data.

Challenges:

- Identifying subtle malicious patterns in large datasets.
- Differentiating between legitimate outliers and poisoned data points.
- Maintaining model performance after removing suspicious data.

INTRUSION
SECOND EDITION



AI/ML

9. AI-Powered Multi-Modal Deepfake Detection and Authentication System

Problem Statement:

Develop a system capable of detecting deepfake images, videos, and audio using AI and verifying the authenticity of digital media in real time.

Description:

Deepfake technologies can generate highly realistic fake media that can be used for misinformation, fraud, and identity impersonation. An AI-based system can analyze visual, audio, and metadata signals to detect manipulated media and verify authenticity.

Challenges:

- Detecting deepfakes across images, videos, and audio.
- Identifying subtle inconsistencies like facial expressions and audio-visual mismatches.
- Supporting real-time detection with high accuracy.

INTRUSION
SECOND EDITION



AI/ML

10. AI-Powered Data Loss Prevention for Enterprise Systems

Problem Statement:

Develop an AI system capable of detecting and preventing sensitive data leakage across enterprise networks.

Description:

Organizations store sensitive information such as personal data, financial records, and intellectual property. AI-based systems can analyze network traffic, files, and user activity to detect potential data leaks and prevent unauthorized data transfers.

Challenges:

- Detecting sophisticated data exfiltration techniques.
- Analyzing large volumes of network and user activity data.
- Preventing data leaks while minimizing false positives.

INTRUSION
SECOND EDITION

AI/ML

11. Network-Level Data Exfiltration Detection using AI

Problem Statement:

Create a system that detects large-scale data exfiltration attempts from enterprise networks.

Description:

Attackers often steal sensitive data by gradually transferring it outside the organization to avoid detection. An AI-based monitoring system can analyze network traffic patterns and identify suspicious data transfer activities.

Challenges:

- Detecting slow and stealthy data exfiltration attempts.
- Analyzing large volumes of network traffic efficiently.
- Minimizing false positives while maintaining accurate detection.

INTRUSION
SECOND EDITION



AI/ML

12. Hardening Command & Control (C2) Against Synthetic Identity Injection

Problem Statement:

Develop a system that prevents deepfake-based impersonation attacks targeting command and control systems.

Description:

Advancements in generative AI enable attackers to create deepfake audio and video to impersonate officials and issue fraudulent commands. A secure system is required to verify the authenticity of communications and prevent such manipulation.

Challenges:

- Detecting deepfake audio and video in real time.
- Building resilience against adversarial manipulation techniques.
- Ensuring secure media authenticity verification mechanisms.

INTRUSION
SECOND EDITION



AI/ML

13. AI-Powered Mobile Security & Awareness Companion

Problem Statement:

Mobile users often fall victim to malicious links and social engineering attacks through SMS and messaging applications due to limited awareness of cybersecurity threats.

Description:

Develop an AI-based Android application that acts as a personal mobile security assistant. The app should analyze incoming messages in real time to detect phishing links, malware URLs, or manipulative language. It should also provide users with updated information about cybersecurity threats and offer guidance on protecting their digital identity.

Challenges:

- Accurately detecting phishing links and social engineering messages in real time.
- Providing reliable and up-to-date cybersecurity news and threat intelligence.
- Designing an easy-to-use interface that improves user awareness and security practices.

INTRUSION
SECOND EDITION



AI/ML

14. Deepfake Voice & Video Scam Detector

Problem Statement:

Scammers increasingly use AI to clone voices or generate fake videos of family members or executives to deceive victims into sending money or sensitive information.

Description:

Develop a lightweight AI tool (browser extension or mobile application) that analyzes audio or video content in real time or from uploaded files to detect AI-generated media. The system should examine speech patterns, audio frequencies, and visual cues to identify signs of synthetic generation and warn users about potential scams.

Challenges:

- Detecting subtle patterns in AI-generated audio and video.
- Providing accurate real-time analysis with minimal processing overhead.
- Building a reliable reporting system to share and track known scam profiles or numbers.

INTRUSION
SECOND EDITION



INTRUSION

SECOND EDITION

WEB & APP

WEB & APP

1. Automated Vulnerability Monitoring Tool

Problem Statement:

Organizations need an automated way to track newly published security vulnerabilities affecting their systems.

Description:

A web-based tool that scans vendor and security websites to collect information about newly disclosed vulnerabilities. The system should extract details such as CVE IDs, severity levels, and mitigation steps to help organizations stay updated and secure their systems.

Challenges:

- Handling frequently changing website formats.
- Avoiding duplicate vulnerability reports.
- Managing and processing large volumes of vulnerability data efficiently.

INTRUSION
SECOND EDITION



WEB & APP

2. Unencrypted PII in Browser Storage & Debug Logs

Problem Statement:

Sensitive user information may be unintentionally exposed in browser storage and debug logs of web applications.

Description:

Web applications sometimes store sensitive data such as credit card numbers, authentication tokens, or personal details in browser storage (localStorage or sessionStorage) or print them in debug logs. This data can be accessed by malicious scripts or browser extensions, leading to data leakage.

Challenges:

- Identifying different categories of leaked PII in browser logs and storage.
- Demonstrating how attacks like XSS can exploit this exposed data.
- Implementing security measures such as data redaction, secure storage practices, and strict content security policies.

INTRUSION
SECOND EDITION



WEB & APP

3. Centralized Application Firewall System

Problem Statement:

Organizations need a centralized firewall system to monitor and control application network communication.

Description:

An application-aware firewall that allows administrators to define and manage security policies for domains, IP addresses, and communication protocols through a centralized dashboard. It helps monitor traffic and enforce security rules across multiple applications.

Challenges:

- Managing security policies across multiple endpoints.
- Detecting abnormal or malicious network activities.
- Ensuring minimal impact on application performance.

INTRUSION
SECOND EDITION



WEB & APP

4. AI-Based Web Application Attack Detection

Problem Statement:

Web applications need intelligent systems to automatically detect and prevent cyberattacks.

Description:

An AI-driven system that analyzes HTTP requests in real time to detect and block attacks such as SQL Injection, Cross-Site Scripting (XSS), and Cross-Site Request Forgery (CSRF). The system may also include vulnerability scanning features for developers.

Challenges:

- Minimizing false positives in attack detection.
- Handling large volumes of web traffic efficiently.
- Providing useful security insights to developers and administrators.

INTRUSION
SECOND EDITION



WEB & APP

5. Secure Authentication and Session Management System

Problem Statement:

Build a secure authentication framework that protects web applications from session hijacking and credential-based attacks.

Description:

Weak authentication mechanisms and poor session management can allow attackers to bypass login systems or hijack active user sessions. A secure system should implement strong authentication methods, secure session handling, and monitoring mechanisms to detect suspicious login activities.

Challenges:

- Preventing session hijacking and credential theft.
- Implementing strong authentication mechanisms.
- Detecting and responding to abnormal login behavior.

INTRUSION
SECOND EDITION



WEB & APP

6. Web Application Supply Chain Security Scanner

Problem Statement:

Develop a tool that scans web applications for vulnerable third-party libraries and dependencies.

Description:

Modern web applications depend heavily on external packages and libraries. These dependencies may contain known security vulnerabilities that attackers can exploit. The system should analyze application dependencies and alert developers about vulnerable libraries with recommended fixes.

Challenges:

- Detecting vulnerabilities in third-party dependencies.
- Continuously updating vulnerability databases.
- Providing accurate alerts and remediation suggestions to developers.

INTRUSION
SECOND EDITION



WEB & APP

7. Secure API Gateway for Web Applications

Problem Statement:

Create a secure gateway that protects web APIs from unauthorized access and abuse.

Description:

Modern applications rely heavily on APIs to exchange data between services. If APIs are not properly secured, they can expose sensitive information or allow attackers to exploit system functionality. A secure API gateway should enforce authentication, rate limiting, request monitoring, and logging to ensure safe API communication.

Challenges:

- Preventing unauthorized API access.
- Implementing effective rate limiting and abuse detection.
- Maintaining high performance while enforcing security controls.

INTRUSION
SECOND EDITION



WEB & APP

8. GraphQL Introspection & Batching Attack for DoS

Problem Statement

GraphQL APIs may expose introspection features and batching capabilities that attackers can exploit to perform denial-of-service (DoS) attacks.

Description:

GraphQL introspection allows attackers to explore the entire API schema, including queries and mutations. Attackers can then craft deeply nested queries or batch multiple operations in a single request, exhausting server resources such as CPU, memory, or database connections.

Challenges:

- Detecting high-complexity and deeply nested queries.
- Preventing excessive batched operations in a single request.
- Implementing query depth limits, complexity analysis, and rate limiting to prevent resource exhaustion.

INTRUSION
SECOND EDITION

WEB & APP

9. HTTP Request Smuggling in Reverse-Proxy Chains

Problem Statement:

HTTP Request Smuggling exploits inconsistencies in how front-end proxies and back-end servers interpret HTTP headers, allowing attackers to manipulate request processing.

Description:

When proxies (like Nginx or HAProxy) and backend servers parse HTTP headers such as Content-Length and Transfer-Encoding differently, attackers can craft malformed requests. This can cause one server to treat part of a request differently than another, enabling attacks like session hijacking, cache poisoning, or bypassing security controls.

Challenges:

- Identifying discrepancies in request parsing between proxies and backend servers.
- Demonstrating request queue poisoning through crafted payloads.
- Implementing defenses such as header normalization, strict protocol validation, and secure proxy configurations.

INTRUSION
SECOND EDITION



WEB & APP

10. Digital Intelligence Aggregation Platform

Problem Statement

Develop a Digital Intelligence Aggregation Platform that combines multiple OSINT capabilities into a single system, enabling investigators to collect and analyze intelligence from publicly available digital sources through a unified interface.

Description

Investigators often rely on multiple OSINT tools to gather information from open sources such as social media, domain records, public databases, and network intelligence services. This platform should provide a centralized dashboard where users can perform tasks like username searches, email analysis, domain and IP lookup, metadata extraction, and geolocation tracking.

Challenges

- Building OSINT collection modules from scratch without relying on external APIs.
- Aggregating and organizing data from multiple open sources.
- Handling large volumes of intelligence while maintaining performance and accuracy.

INTRUSION
SECOND EDITION



INTRUSION

SECOND EDITION

CYBERSECURITY



CYBERSECURITY

1. Early Warning System for Android Ransomware

Problem Statement:

There is a need for an early warning system to detect ransomware activity on Android devices before major damage occurs.

Description:

Android ransomware can encrypt files or lock screens, threatening personal and financial data. A behavior-based system can monitor suspicious actions such as rapid file encryption or lockscreen hijacking and provide real-time alerts to the user.

Challenges:

- Designing a lightweight system that does not affect device performance.
- Implementing accurate alerts with low false positives.
- Developing preventive mechanisms such as warnings or rollback options.

INTRUSION
SECOND EDITION



CYBERSECURITY

2. Analysis of Tor Network Hidden Services

Problem Statement:

Active hidden services on the Tor network hosting malicious content are difficult to discover and analyze efficiently.

Description:

Tor allows anonymous hosting through .onion websites, which can sometimes be used for illegal activities. The goal is to build a system that identifies active hidden services and analyzes them for vulnerabilities or malicious behavior while maintaining ethical and legal practices.

Challenges:

- Discovering active .onion services without compromising anonymity.
- Analyzing potential vulnerabilities in hidden services.
- Handling large-scale data collection while ensuring legal and ethical compliance.

INTRUSION
SECOND EDITION



CYBERSECURITY

3. Secure DNS Filtering System

Problem Statement:

Malicious domains used by malware and botnets often evade traditional DNS security mechanisms.

Description:

Attackers frequently use DNS to communicate with command-and-control (C2) servers. This system aims to build a secure DNS filtering solution that detects and blocks malicious domains in real time using threat intelligence and anomaly detection techniques.

Challenges:

- Detecting domains generated by Domain Generation Algorithms (DGA).
- Identifying DNS tunneling attempts over modern protocols like DoH and DoT.
- Maintaining fast DNS resolution with minimal latency.

INTRUSION
SECOND EDITION



CYBERSECURITY

4. Centralized Infrastructure Management Platform

Problem Statement:

Manual and fragmented management of servers and network infrastructure creates operational inefficiencies and security risks.

Description:

Managing distributed servers, firewalls, and infrastructure using multiple tools increases the risk of misconfiguration. The goal is to build a centralized platform that allows administrators to securely monitor, manage, and control infrastructure components from a single interface.

Challenges:

- Designing a portal with role-based access control.
- Implementing real-time monitoring and audit logging.
- Supporting secure patching, updates, and rollback mechanisms.

INTRUSION
SECOND EDITION



CYBERSECURITY

5. Automated Fuzzing for IoT Protocols

Problem Statement:

There is a need for automated fuzzing tools to discover vulnerabilities in wireless communication protocols used in IoT systems.

Description:

Communication protocols such as Wi-Fi, Bluetooth, and Zigbee may contain vulnerabilities in their handshake processes or state transitions. An automated fuzzing framework can generate malformed or unexpected inputs to test these protocols and identify potential weaknesses in their communication stacks.

Challenges:

- Implementing fuzzing techniques for protocols like WPA2/WPA3 and BLE.
- Detecting crashes and identifying unique vulnerabilities automatically.
- Designing a modular framework that can support multiple IoT protocols.

INTRUSION
SECOND EDITION

CYBERSECURITY

6. Dark Web Monitoring and Keyword Search Tool

Problem Statement:

There is a need for a centralized tool to monitor illegal activities across dark web platforms using keyword-based search.

Description:

Anonymous networks such as TOR, I2P, and Freenet host platforms that may contain illegal marketplaces or malicious activities. The system should collect and filter data from these networks based on keywords, locations, and timestamps to help investigators monitor suspicious activities.

Challenges:

- Establishing secure connectivity through VPN or anonymity networks.
- Extracting data from dynamic and frequently changing dark web sources.
- Managing continuously changing dark web resources while maintaining operational security.

INTRUSION
SECOND EDITION



CYBERSECURITY

7. Cybersecurity Readiness Assessment Tool

Problem Statement:

Organizations lack structured tools to assess their cybersecurity readiness against modern threats.

Description:

Many organizations struggle to evaluate how prepared they are for threats such as malware, phishing, and data breaches. This tool will assess an organization's capabilities across areas like prevention, detection, response, and recovery, and generate a cybersecurity maturity score with recommendations.

Challenges:

- Designing a reliable assessment framework.
- Generating meaningful maturity scores and actionable recommendations.
- Ensuring the confidentiality and security of assessment data.

INTRUSION
SECOND EDITION



CYBERSECURITY

8. BGP Route Hijack Detection & RPKI Enforcement System

Problem Statement

Develop a system to detect BGP prefix hijacking, where unauthorized autonomous systems announce IP prefixes they do not own, potentially causing traffic diversion, interception, or service disruption.

Description

Build a real-time BGP monitoring system that analyzes live BGP update streams (RouteViews or RIPE RIS) to detect routing anomalies. The system should validate prefix announcements against RPKI ROA records and identify suspicious behavior such as origin AS mismatches, unauthorized subprefix announcements, and multi-origin routes. Detected events should generate alerts and be displayed on a simple monitoring dashboard.

Challenges

- Processing live BGP update streams efficiently.
- Validating prefix-to-origin mappings using RPKI data.
- Detecting hijacks through routing anomaly heuristics.
- Designing a clear alerting and monitoring dashboard.

INTRUSION
SECOND EDITION



CYBERSECURITY

9. Cryptographic Agility Framework for Post-Quantum Migration

Problem Statement

Develop a framework that enables enterprises to transition from classical cryptography (RSA, ECDH, ECDSA) to post-quantum cryptography (PQC) while maintaining compatibility and security during the migration process.

Description

The system should support cryptographic agility, allowing algorithms to be replaced without modifying application code. It will identify cryptographic usage in a codebase, classify algorithms based on quantum vulnerability, and provide a unified cryptographic interface supporting classical, post-quantum, and hybrid implementations. This ensures secure migration to NIST PQC standards such as ML-KEM and ML-DSA.

Challenges

- Identifying and classifying cryptographic algorithms within existing codebases.
- Designing a unified crypto API that supports classical, PQC, and hybrid backends.
- Ensuring compatibility during migration between classical and post-quantum systems.
- Testing secure key exchange, encryption, and signature operations across modes.

INTRUSION
SECOND EDITION



INTRUSION

SECOND EDITION

CLOUD



@official_cyberonites

CLOUD

1. Software-Defined Load Balancer, Firewall, and WAF for Cloud-Native Applications

Problem Statement:

Design and integrate a software-defined load balancer, firewall, and web application firewall (WAF) for cloud-native application stacks.

Description:

Modern cloud environments use software-defined networking to manage traffic and security. This project involves selecting suitable open-source solutions and integrating them with orchestration platforms such as Kubernetes or OpenStack. The system should support load balancing, SSL offloading, and WAF protection using tools like Nginx or Apache with security rules.

Challenges:

- Selecting and integrating suitable open-source security solutions.
- Implementing WAF rules to protect applications from web attacks.
- Ensuring seamless integration with cloud orchestration platforms.

INTRUSION
SECOND EDITION

CLOUD

2. Unified Monitoring Solution for VMs and Containers

Problem Statement:

Create a unified monitoring system for both virtual machines (VMs) and containerized environments.

Description:

Modern cloud infrastructures run workloads across both virtual machines and containers. This project involves building a monitoring platform that collects performance metrics from different environments, stores them in a real-time database, and provides visualization dashboards along with alerting capabilities.

Challenges:

- Collecting metrics from diverse environments (VMs and containers).
- Efficiently storing large volumes of monitoring data.
- Designing dashboards and alert mechanisms for real-time monitoring.

INTRUSION
SECOND EDITION



CLOUD

3. Cloud Identity and Access Mismanagement Detection Tool

Problem Statement:

Improper Identity and Access Management (IAM) policies in cloud environments can allow privilege escalation and unauthorized access.

Description:

Cloud platforms rely heavily on IAM policies to control user permissions and access to resources. Misconfigured or overly permissive IAM policies can lead to serious security risks. The goal is to build a tool that analyzes IAM configurations and identifies excessive, risky, or misconfigured permissions.

Challenges:

- Detecting overly permissive IAM policies.
- Identifying potential privilege escalation paths.
- Providing clear remediation recommendations for secure access management.

INTRUSION
SECOND EDITION

CLOUD

4. Cloud-Based Satellite Data Processing System

Problem Statement:

Build a scalable cloud system capable of processing large volumes of satellite and geospatial data.

Description:

Satellite missions and earth observation programs generate massive datasets that need efficient storage, processing, and analysis. The system should provide a scalable cloud platform where researchers can process, analyze, and visualize geospatial data efficiently.

Challenges:

- Managing and processing very large geospatial datasets.
- Ensuring scalability and performance in cloud environments.
- Providing effective visualization and analysis tools for researchers.

INTRUSION
SECOND EDITION

CLOUD

5. DNS Rebinding Attack on Internal Admin Panels

Problem Statement:

DNS rebinding attacks can bypass browser security policies and allow attackers to access internal services such as admin panels and local servers.

Description:

In a DNS rebinding attack, an attacker controls a domain that initially resolves to a legitimate IP address but later changes to an internal IP address. This tricks the victim's browser into sending requests to internal systems, potentially exposing sensitive data or configurations.

Challenges:

- Simulating a DNS rebinding attack in a controlled environment.
- Demonstrating data exfiltration from internal services.
- Implementing defenses such as host validation, network access policies, and secure service binding.

INTRUSION
SECOND EDITION



CLOUD

6. Insecure Serverless Applications

Problem Statement:

Serverless applications may contain security vulnerabilities due to insecure dependencies, improper permissions, or weak configurations.

Description:

Serverless computing platforms such as AWS Lambda and Azure Functions remove the need for infrastructure management but introduce new security risks. The goal is to develop methods or tools that detect vulnerabilities in serverless functions and help developers mitigate these risks.

Challenges:

- Identifying insecure dependencies in serverless functions.
- Detecting improper permissions and configuration issues.
- Providing automated security checks without affecting application performance.

INTRUSION
SECOND EDITION



CLOUD

7. Kubernetes RBAC Privilege Escalation via Wildcard Permissions

Problem Statement

Identify and mitigate Kubernetes RBAC misconfigurations involving wildcard (*) permissions that may grant excessive privileges to service accounts and increase the risk of unauthorized access within the cluster.

Description

Kubernetes RBAC controls access to cluster resources, but the use of wildcard permissions in roles or cluster roles can unintentionally grant broad access. This platform focuses on auditing RBAC configurations to detect overly permissive service accounts attached to application pods. It emphasizes strengthening security by implementing least-privilege access controls, continuous RBAC monitoring, and policy enforcement to prevent misuse of elevated permissions.

Challenges

- Detecting wildcard or overly permissive RBAC rules in cluster configurations.
- Auditing service accounts and role bindings for excessive privileges.
- Enforcing least-privilege RBAC policies and automated compliance checks using security tools and policy frameworks.

INTRUSION
SECOND EDITION



CLOUD

8. Data Encryption and Key Management Issues

Problem Statement:

Improper encryption key management in cloud systems can expose sensitive data even when encryption is used.

Description:

Cloud environments rely on encryption to protect stored and transmitted data. However, if encryption keys are poorly managed, leaked, or improperly stored, attackers can still gain access to encrypted information. The goal is to design a secure system for managing encryption keys and protecting them from unauthorized access.

Challenges:

- Securing storage and lifecycle management of encryption keys.
- Preventing unauthorized access or leakage of keys.
- Implementing scalable key management mechanisms for cloud environments.

INTRUSION
SECOND EDITION





INTRUSION

SECOND EDITION

BLOCK CHAIN



@official_cyberonites

BLOCKCHAIN

1. Sybil Attacks on Decentralized Identity (DID) Systems

Problem Statement:

Decentralized Identity (DID) systems on blockchain are vulnerable to Sybil attacks, where attackers create multiple fake identities to manipulate the network.

Description:

In Industrial IoT (IIoT) blockchain networks, decentralized identities are used for device authentication and access control. Attackers can create thousands of fake identities to influence consensus, corrupt voting mechanisms, or bypass access controls in the network.

Challenges:

- Detecting and eliminating Sybil nodes without relying on centralized authorities.
- Maintaining privacy and anonymity of legitimate devices.
- Identifying fraudulent identities in real time without disrupting the network.

INTRUSION
SECOND EDITION



BLOCKCHAIN

2. Quantum Threats to Blockchain Cryptography

Problem Statement:

Blockchain transaction data secured by current cryptographic methods may become vulnerable to attacks from quantum computers.

Description:

Most blockchain systems rely on elliptic curve cryptography (ECC) for securing transactions and digital signatures. Future quantum computers could potentially break these cryptographic algorithms, threatening the security and integrity of decentralized networks.

Challenges:

- Migrating existing blockchain systems to quantum-resistant cryptographic algorithms.
- Maintaining network performance and decentralization during the transition.
- Implementing post-quantum cryptography at large scale without disrupting operations.

INTRUSION
SECOND EDITION

BLOCKCHAIN

3. Zero-Day Vulnerabilities in IIoT Smart Contracts

Problem Statement:

Smart contracts in Industrial IoT (IIoT) blockchain networks may contain undiscovered zero-day vulnerabilities that can be exploited by attackers.

Description:

Once deployed, smart contracts cannot easily be modified, making any hidden vulnerability a permanent attack surface. Attackers can exploit these vulnerabilities to manipulate transactions, disrupt industrial processes, or cause financial and operational damage.

Challenges:

- Performing automated security auditing of smart contract code.
- Detecting malicious contract interactions in real time.
- Maintaining decentralization while monitoring and preventing attacks.

INTRUSION
SECOND EDITION



BLOCKCHAIN

4. Cross-Chain Bridge Trust Gaps

Problem Statement:

Cross-chain bridges used in multi-blockchain systems may introduce trust gaps that attackers can exploit to manipulate data.

Description:

Industrial IoT ecosystems may use multiple blockchain networks that communicate through cross-chain bridges. During this transfer, cryptographic continuity can break, allowing attackers to inject falsified device telemetry or manipulate data as it moves between blockchains.

Challenges:

- Maintaining end-to-end cryptographic proof across different blockchain networks.
- Detecting anomalies during cross-chain data transfers.
- Preventing centralized bridge validators that could become single points of failure.

INTRUSION
SECOND EDITION



BLOCKCHAIN

5. Detecting 51% Attacks on Critical Infrastructure Audit Logs

Problem Statement:

Blockchain-based audit logs used in critical infrastructure can be manipulated through 51% attacks, allowing attackers to rewrite recent transaction history.

Description:

Industries such as power grids and water treatment systems use blockchain for tamper-proof logging. However, if attackers gain control of more than 50% of the network's mining power, they can alter recent blocks, erase evidence of cyberattacks, and falsify sensor data records.

Challenges:

- Detecting a 51% attack before major chain reorganization occurs.
- Protecting audit logs from manipulation in low-node blockchain networks.
- Designing consensus mechanisms that resist majority attacks.

INTRUSION
SECOND EDITION



INTRUSION

SECOND EDITION

IOT



@official_cyberonites

IOT

1. IoT Botnet Detection System

Problem Statement:

Design a real-time detection system that identifies IoT devices infected by botnets participating in large-scale DDoS attacks.

Description:

IoT botnets such as Mirai infect devices like routers, cameras, and smart appliances and use them to launch distributed denial-of-service (DDoS) attacks. The system should monitor network traffic to detect suspicious activity and automatically isolate compromised devices without disrupting legitimate network operations.

Challenges:

- Detecting abnormal traffic patterns from IoT devices.
- Differentiating between legitimate and malicious network activity.
- Isolating infected devices without affecting normal network functionality.

INTRUSION
SECOND EDITION

IoT

2. Smart Home IoT Intrusion Detection System

Problem Statement:

Design a home network monitoring system that detects compromised IoT devices such as cameras, routers, or smart speakers.

Description:

Smart home devices often lack strong security and can be targeted by attackers for data theft or device control. The system should monitor network behavior of IoT devices and alert users if suspicious activity or unauthorized access is detected.

Challenges:

- Monitoring diverse IoT devices with different communication protocols.
- Detecting abnormal behavior indicating device compromise.
- Providing timely alerts without generating excessive false positives.

INTRUSION
SECOND EDITION



3. Firmware Integrity Verification System for Industrial IoT

Problem Statement:

Design a lightweight, offline-capable system that verifies firmware integrity in Industrial IoT mesh networks to prevent malicious OTA firmware injection.

Description:

Industrial IoT devices receive firmware updates through Over-The-Air (OTA) mechanisms. Attackers can perform man-in-the-middle attacks to inject malicious firmware that manipulates sensor readings or disables safety systems. The system should verify firmware authenticity before installation and ensure updates have not been tampered with.

Challenges:

- Implementing firmware verification within limited memory and processing capacity of IoT devices.
- Designing reliable peer verification even if some nodes are compromised.
- Ensuring update verification does not delay legitimate firmware deployment.

INTRUSION
SECOND EDITION



4. Passive Network Traffic Analysis for Compromised IoT Devices

Problem Statement:

Design a passive network monitoring system that detects IoT devices compromised through supply-chain attacks by analyzing their network behavior.

Description:

Many IoT devices may ship with malicious firmware that secretly communicates with remote command-and-control (C2) servers. These communications often appear similar to normal traffic. The system should analyze network behavior patterns to identify hidden C2 activity, suspicious DNS requests, and abnormal outbound connections without accessing the device itself.

Challenges:

- Detecting hidden C2 communication patterns in encrypted traffic.
- Distinguishing between normal device behavior and malicious activity.
- Building reliable behavioral models with low false positives.

INTRUSION
SECOND EDITION

5. Healthcare IoT Security Monitoring System

Problem Statement:

Build a security platform that monitors hospital IoT devices and detects vulnerabilities or suspicious communication in real time.

Description:

Hospitals use IoT-enabled medical devices such as ventilators, infusion pumps, and imaging systems. These devices may have security vulnerabilities that attackers could exploit. The system should monitor device behavior and network communication to identify potential threats and alert administrators.

Challenges:

- Monitoring diverse medical IoT devices securely.
- Detecting vulnerabilities and abnormal device communication.
- Ensuring patient safety while implementing security monitoring.

INTRUSION
SECOND EDITION

6. Autonomous IoT Threat Response System

Problem Statement:

Create an automated system that detects compromised IoT endpoints and isolates them instantly while preserving critical system functionality.

Description:

IoT devices in enterprise or critical environments may become compromised by malware or remote attackers. An autonomous threat response system should continuously monitor device behavior, detect signs of compromise, and automatically isolate or contain affected devices to prevent the spread of attacks while keeping essential services operational.

Challenges:

- Detecting compromised IoT devices in real time.
- Automatically isolating infected devices without disrupting critical operations.
- Ensuring the response system acts quickly while minimizing false positives.

INTRUSION
SECOND EDITION



7. Insecure Smart Home Cameras (Doorbells).

Problem Statement:

Many smart home cameras and doorbells are deployed with weak or default passwords and poor security configurations. Attackers can exploit these vulnerabilities to gain unauthorized access to live video feeds and microphones, leading to privacy breaches and potential harassment of users.

Description:

Weak authentication, insecure device configurations, and lack of encryption allow attackers to compromise IoT cameras through brute-force or credential attacks. Once accessed, attackers can monitor homes or misuse the devices for malicious purposes.

Challenges:

- Default/weak credentials on cameras (unchanged from factory).
- Lack of strong authentication (no 2FA, insecure resets).
- Unencrypted video/audio streams.
- User complacency about IoT device security.

INTRUSION
SECOND EDITION

8. Vulnerable Smart Building Access Controls

Problem Statement:

Smart building access systems such as IoT-based door locks are increasingly used in offices. Security flaws in these systems can allow attackers to bypass authentication, manipulate credentials, and remotely unlock doors, creating serious physical and cybersecurity risks.

Description:

Vulnerabilities in smart lock systems can enable attackers to gain unauthorized physical access, deploy malware, or disrupt access control services. Weak security practices and delayed patching make these cyber-physical systems attractive targets for attackers.

Challenges:

- Unauthorized physical access due to system compromise.
- Unpatched firmware and delayed security updates.
- Default or predictable device credentials.
- Difficulty securing cyber-physical access control systems.

INTRUSION
SECOND EDITION

